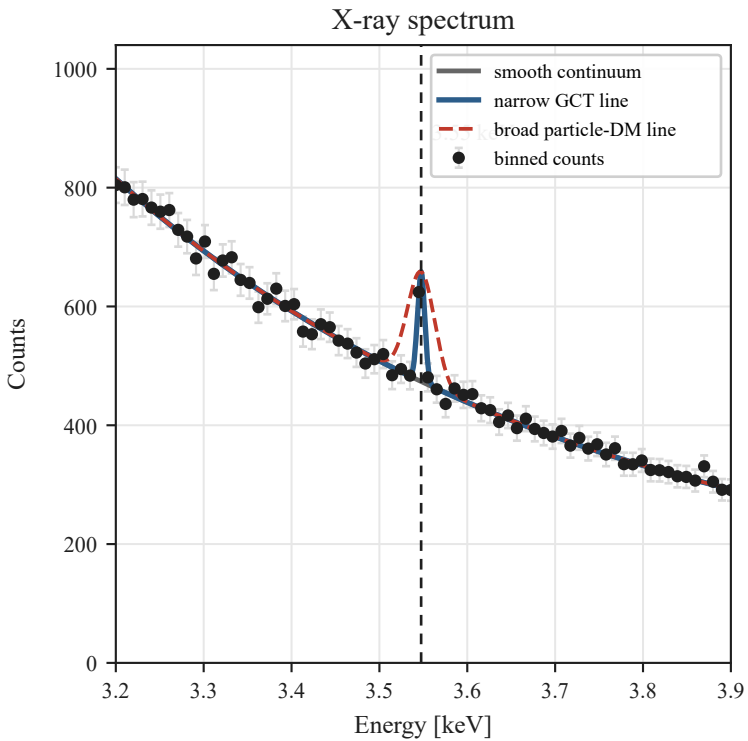
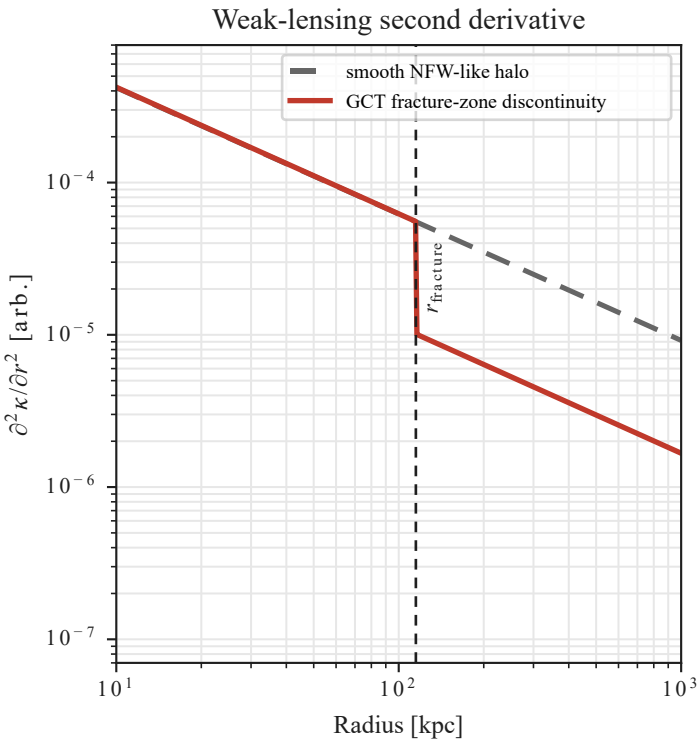
The 3.55 keV X-ray Anomaly: Detection and Morphology Tests

Perseus cluster context: line shape plus second-derivative lensing morphology.





XRISM line gate

GCT FWHM ~ 9.7 eV
CDM FWHM ~ 36.6 eV

Morphology gate

sharp lensing second-derivative feature

Null outcome

smooth halo + no narrow residual

Observation: line residual + lensing map → narrow line + discontinuity supports fracture morphology → broad line or smooth halo routes to particle-DM/null